

Mathematics for Physics 2 — Week 1 Review

Sources: *Memphis_2_week_1.pdf*, *ClassEx1 (2026).pdf*

Pending approval before writing to cheat sheet

Legend: **INCLUDE** — will be added to cheat sheet **EXCLUDE** — excluded **BORDER-LINE** — needs your decision

Notation & Definitions

#	Formula / Definition	Decision	Reason
N1	Hat notation: $\hat{v}$ denotes a <i>unit vector</i> in the direction of $\vec{v}$: $\hat{v} = \frac{\vec{v}}{ \vec{v} }, \quad \hat{v} = 1$	INCLUDE	Foundational notation
N2	Vector components: A vector $\vec{A}$ in 3D Cartesian coordinates is written $\vec{A} = A_x \hat{x} + A_y \hat{y} + A_z \hat{z}$ A_i denotes the i -th component of $\vec{A}$ (e.g. $A_1 = A_x$). The magnitude is $ \vec{A} = A = \sqrt{A_x^2 + A_y^2 + A_z^2}$.	INCLUDE	Required to understand index notation

Index Notation & Einstein Summation

#	Formula	Decision	Reason
1	$\vec{A} = A_i \hat{e}_i$ (sum over $i = 1, 2, 3$ implied)	INCLUDE	Core notation
2	Einstein convention: any index repeated <i>exactly twice</i> in a term implies summation over 1, 2, 3. Example: $a_i b_i \equiv \sum_{i=1}^3 a_i b_i$	INCLUDE	Fundamental rule
3	Index conservation: free indices must be the same on both sides of an equation.	INCLUDE	Key consistency rule

Kronecker Delta

#	Formula	Decision	Reason
4	$\delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$	INCLUDE	Definition
5	$\delta_{ij} = \delta_{ji}$	INCLUDE	Symmetry
6	$A_j \delta_{ij} = A_i$	INCLUDE	Index contraction — used constantly
7	$\delta_{ik} \delta_{jk} = \delta_{ij}$	INCLUDE	Product identity
(was $\delta_{ii} = 3$ 8)		EXCLUDE	Dropped per review

Levi-Civita Symbol

#	Formula	Decision	Reason
10	$\varepsilon_{ijk} = \begin{cases} +1 & (i, j, k) \text{ cyclic: } (1, 2, 3), (2, 3, 1), (3, 1, 2) \\ -1 & \text{anti-cyclic: } (2, 1, 3), (1, 3, 2), (3, 2, 1) \\ 0 & \text{any two indices equal} \end{cases}$	INCLUDE	Definition
11	$\varepsilon_{ijk} = -\varepsilon_{jik}$ (swapping any two adjacent indices flips sign)	INCLUDE	Key property
(was $\varepsilon_{ijk} = \varepsilon_{kij} = \varepsilon_{jki}$ 12)		EXCLUDE	Dropped per review
13	$\sum_k \varepsilon_{ijk} \varepsilon_{mnk} = \delta_{im} \delta_{jn} - \delta_{in} \delta_{jm}$	INCLUDE	Simplifies double cross products

Dot Product (Scalar Product)

#	Formula	Decision	Reason
14	$\vec{A} \cdot \vec{B} = AB \cos \theta$ (θ = angle between $\vec{A}$ and $\vec{B}$)	INCLUDE	Geometric definition
15	$\vec{A} \cdot \vec{B} = A_i B_i = \sum_{i=1}^3 A_i B_i$	INCLUDE	Index form
16	$\hat{e}_i \cdot \hat{e}_j = \delta_{ij}$	INCLUDE	Orthonormality
17	$ \vec{A} = \sqrt{\vec{A} \cdot \vec{A}} = \sqrt{A_i A_i} = \sqrt{\sum_{i=1}^3 A_i^2}$ Note: $A_i A_i$ uses Einstein convention — i appears twice, so it sums: $A_1^2 + A_2^2 + A_3^2$.	INCLUDE	Clarified
18	$\vec{A} \cdot \vec{B} = 0 \Rightarrow \vec{A} \perp \vec{B}$	EXCLUDE	Trivially obvious from #14
19	$\vec{A} \cdot \vec{B} = AB$ when parallel	EXCLUDE	Special case of #14
20	$\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$	EXCLUDE	Obvious from $A_i B_i = B_i A_i$

Cross Product (Vector Product)

#	Formula	Decision	Reason
21	$ \vec{a} \times \vec{b} = ab \sin \theta$ (θ = angle between $\vec{a}$ and $\vec{b}$)	INCLUDE	Geometric definition
22	If $\vec{C} = \vec{A} \times \vec{B}$, then the i -th component of $\vec{C}$ is: $C_i = \varepsilon_{ijk} A_j B_k$ (j and k are summed; i is the free index selecting the component)	INCLUDE	Clarified
23	$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}$	INCLUDE	Practical computation formula
24	$(a_y b_z - a_z b_y)\hat{x} + (a_z b_x - a_x b_z)\hat{y} + (a_x b_y - a_y b_x)\hat{z}$	EXCLUDE	Derivable from #23
25	$\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$	INCLUDE	Anti-commutativity
26	$\hat{x} \times \hat{y} = \hat{z}, \quad \hat{y} \times \hat{z} = \hat{x}, \quad \hat{z} \times \hat{x} = \hat{y}$ $\hat{y} \times \hat{x} = -\hat{z}, \quad \hat{z} \times \hat{y} = -\hat{x}, \quad \hat{x} \times \hat{z} = -\hat{y}$ $\hat{x} \times \hat{x} = \hat{y} \times \hat{y} = \hat{z} \times \hat{z} = \vec{0}$	INCLUDE	Quick calculation
27	$\vec{a} \cdot (\vec{a} \times \vec{b}) = 0$	EXCLUDE	Consequence of ε

Key Vector Identity

#	Formula	Decision	Reason
28	BAC-CAB rule: $\vec{A} \times (\vec{B} \times \vec{C}) = \vec{B}(\vec{A} \cdot \vec{C}) - \vec{C}(\vec{A} \cdot \vec{B})$ Here $(\vec{A} \cdot \vec{C})$ and $(\vec{A} \cdot \vec{B})$ are <i>scalars</i> (plain numbers); multiplying a scalar by a vector $\vec{B}$ or $\vec{C}$ scales its magnitude.	INCLUDE	Clarified

Coordinate Systems

Polar Coordinates (r, θ)

#	Formula	Decision	Reason
29	$x = r \cos \theta, \quad y = r \sin \theta$ $r = \sqrt{x^2 + y^2}, \quad \theta = \arctan\left(\frac{y}{x}\right)$	INCLUDE	Conversion formulas
30	$\hat{r} = \cos \theta \hat{x} + \sin \theta \hat{y}$ $\hat{\theta} = -\sin \theta \hat{x} + \cos \theta \hat{y}$	INCLUDE	Unit vectors in polar
31	$\hat{x} = \cos \theta \hat{r} - \sin \theta \hat{\theta}$ $\hat{y} = \sin \theta \hat{r} + \cos \theta \hat{\theta}$	INCLUDE	Inverse conversion

Spherical Coordinates (r, θ, φ)

#	Formula	Decision	Reason
32	$x = r \sin \theta \cos \varphi, \quad y = r \sin \theta \sin \varphi, \quad z = r \cos \theta$ $r = \sqrt{x^2 + y^2 + z^2}, \quad \varphi = \arctan\left(\frac{y}{x}\right), \quad \theta = \arccos\left(\frac{z}{r}\right)$	INCLUDE	Conversion formulas
33	$\hat{r} = \sin \theta \cos \varphi \hat{x} + \sin \theta \sin \varphi \hat{y} + \cos \theta \hat{z}$ $\hat{\theta} = \cos \theta \cos \varphi \hat{x} + \cos \theta \sin \varphi \hat{y} - \sin \theta \hat{z}$ $\hat{\varphi} = -\sin \varphi \hat{x} + \cos \varphi \hat{y}$	INCLUDE	Unit vectors in spherical
34	$\hat{x} = \sin \theta \cos \varphi \hat{r} + \cos \theta \cos \varphi \hat{\theta} - \sin \varphi \hat{\varphi}$ $\hat{y} = \sin \theta \sin \varphi \hat{r} + \cos \theta \sin \varphi \hat{\theta} + \cos \varphi \hat{\varphi}$ $\hat{z} = \cos \theta \hat{r} - \sin \theta \hat{\theta}$	INCLUDE	Included per review